

SI 140A-01 Probability & Statistics for EECS, Fall 2026

Homework 1

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Read all the instructions below carefully before you start working on the assignment, and before you make a submission.

- You are required to write down all the major steps towards making your conclusions; otherwise you may obtain limited points of the problem.
- Write your homework in English; otherwise you will get no points of this homework.
- Any form of plagiarism will lead to 0 point of this homework.

Problem 1

(Story Proof) Define $\left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}$ as the number of ways to partition $\{1, 2, \dots, n\}$ into k non-empty subsets, or the number of ways to have n students split up into k groups such that each group has at least one student. For example, $\left\{ \begin{smallmatrix} 4 \\ 2 \end{smallmatrix} \right\} = 7$ because we have the following possibilities:

- $\{1\}, \{2, 3, 4\}$ • $\{1, 2\}, \{3, 4\}$
- $\{2\}, \{1, 3, 4\}$ • $\{1, 3\}, \{2, 4\}$
- $\{3\}, \{1, 2, 4\}$ • $\{1, 4\}, \{2, 3\}$
- $\{4\}, \{1, 2, 3\}$

Prove the following identities:

(a)

$$\left\{ \begin{smallmatrix} n+1 \\ k \end{smallmatrix} \right\} = \left\{ \begin{smallmatrix} n \\ k-1 \end{smallmatrix} \right\} + k \left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}.$$

Hint: I'm either in a group by myself or I'm not.

(b)

$$\sum_{j=k}^n \binom{n}{j} \left\{ \begin{smallmatrix} j \\ k \end{smallmatrix} \right\} = \left\{ \begin{smallmatrix} n+1 \\ k+1 \end{smallmatrix} \right\}.$$

Hint: First decide how many people are not going to be in my group.

(a) answer: Assuming there are $n+1$ people (including me) need to be divided into k groups. Special consideration for myself. $\left\{ \begin{smallmatrix} n \\ k-1 \end{smallmatrix} \right\}$ represents the cases that I form a group on my own and the rest people form $k-1$ groups. $k \left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}$ represents the cases that the rest n people form k groups at first and I choose one of them to join. $\left\{ \begin{smallmatrix} n \\ k-1 \end{smallmatrix} \right\} + k \left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}$ represents all the cases (I form a group on my own plus I form a group with other people), which equals to $\left\{ \begin{smallmatrix} n+1 \\ k \end{smallmatrix} \right\}$.

(b) answer: Assuming there are $n+1$ people need to be divided into $k+1$ groups. $\binom{n}{j} \left\{ \begin{smallmatrix} j \\ k \end{smallmatrix} \right\}$ represents choose j people (except me) to form k groups. Me and the rest people are in the last group. Every different j corresponds to different cases. Summing up the cases corresponding to different j values will give the final number of all cases. So the equation

$$\sum_{j=k}^n \binom{n}{j} \left\{ \begin{smallmatrix} j \\ k \end{smallmatrix} \right\} = \left\{ \begin{smallmatrix} n+1 \\ k+1 \end{smallmatrix} \right\}.$$

is proved.

Problem 2

A *norepeatword* is a sequence of at least one (and possibly all) of the usual 26 letters a, b, c, ..., z, with repetitions not allowed. For example, "course" is a norepeatword, but "statistics" is not. Order matters, e.g., "course" is not the same as "source". A norepeatword is chosen randomly, with all norepeatwords equally likely. Show that the probability that it uses all 26 letters is very close to $1/e$.

answer: Assuming the length of the *norepeatword* is n . According to the question, $n \in \{1, 2, \dots, 26\}$. Define the number of words of length n as I_n . And the value of I_n can be calculated in this way:

$$I_n = \binom{26}{n} n! = \frac{26!}{(26-n)!}.$$

The total number of norepeatword is:

$$N = \sum_{i=1}^{26} \frac{26!}{(26-i)!}$$

And I_{26} is:

$$I_{26} = 26!$$

Let event A be defined as choosing norepeatword consisting of 26 letters. Based on the above information, the probability of A can be calculated:

$$P(A) = \frac{I_{26}}{N} = \frac{26!}{\sum_{i=1}^{26} \frac{26!}{(26-i)!}}$$

Divide both the numerator and the denominator by $26!$:

$$P(A) = \frac{1}{1 + \frac{1}{1!} + \frac{1}{2!} + \dots + \frac{1}{25!}}$$

Considering denominator, Let $k = 26 - i$:

$$\sum_{i=1}^{26} \frac{1}{(26-i)!} = \sum_{k=0}^{25} \frac{1}{k!}$$

Based on the infinite series:

$$e = \sum_{x=0}^{\infty} \frac{1}{x!}$$

So that: $\sum_{k=0}^{25} \frac{1}{k!}$ is close to e , $P(A) = \frac{1}{1 + \frac{1}{1!} + \frac{1}{2!} + \dots + \frac{1}{25!}}$ is close to $\frac{1}{e}$.

Therefore, this question is proven.

Problem 3

Given $n \geq 2$ numbers $(a_1, a_2, \dots, a_n)$ with no repetitions, a bootstrap sample is a sequence $(x_1, x_2, \dots, x_n)$ formed from the a_j 's by sampling with replacement with equal probabilities. Bootstrap samples arise in a widely used statistical method known as the bootstrap. For example, if $n = 2$ and $(a_1, a_2) = (3, 1)$, then the possible bootstrap samples are $(3, 3)$, $(3, 1)$, $(1, 3)$, and $(1, 1)$.

(a) How many possible bootstrap samples are there for $(a_1, \dots, a_n)$?

(b) How many possible bootstrap samples are there for $(a_1, \dots, a_n)$, if order does not matter (in the sense that it only matters how many times each a_j was chosen, not the order in which they were chosen)?

(c) One random bootstrap sample is chosen (by sampling from $a_1, \dots, a_n$ with replacement, as described above). Show that not all unordered bootstrap samples (in the sense of (b)) are equally likely. Find an unordered bootstrap sample $\mathbf{b}_1$ that is as likely as possible, and an unordered bootstrap sample $\mathbf{b}_2$ that is as unlikely as possible. Let p_1 be the probability of getting $\mathbf{b}_1$ and p_2 be the probability of getting $\mathbf{b}_2$ (so p_i is the probability of getting the specific unordered bootstrap sample $\mathbf{b}_i$). What is p_1/p_2 ? What is the ratio of the probability of getting an unordered bootstrap sample whose probability is p_1 to the probability of getting an unordered sample whose probability is p_2 ?

(a) answer: Let N_1 be the total number of all the cases:

$$N_1 = n^n$$

(b) answer: Let N_2 be the total number of all the cases. Let x_i represent the number of times that a_i appears ($i \in \mathbb{N}^+$):

$$\begin{aligned} x_1 + x_2 + \dots + x_n &= n \\ x_1, x_2, \dots, x_n &\geq 0 \end{aligned}$$

So that:

$$N_2 = \binom{n+n-1}{n-1} = \binom{2n-1}{n-1}$$

(c) answer: Let x_i represent the number of times that a_i appears ($i \in \mathbb{N}^+$).

Considering one sample $(x_1, x_2, \dots, x_n)$:

Let N_3 represent the number of ordered sequences corresponding to a fixed unordered sample :

$$N_3 = \binom{n}{x_1} \binom{n-x_1}{x_2} \dots \binom{n-x_1-x_2-\dots-x_{n-1}}{x_n} = \frac{n!}{x_1!x_2!\dots x_n!}$$

The probability of each ordered arrangement appearing within this unordered sample is:

$$\frac{1}{n^n}$$

So the probability of the unordered sample $(x_1, \dots, x_n)$ occurring is:

$$P(x_1, \dots, x_n) = \frac{n!}{x_1!x_2!\dots x_n!} \frac{1}{n^n}$$

Since the probability depends on $(x_1, \dots, x_n)$, different unordered samples need not have the same probability.

According to the above content, $\mathbf{b}_1$ is $(a_1, a_2, \dots, a_n)$ and $p_1 = \frac{n!}{n^n}$

$\mathbf{b}_2 = (a_i, a_i, \dots, a_i), i \in \{1, 2, \dots, n\}$. and $p_2 = \frac{1}{n^n}$

$$\frac{p_1}{p_2} = n!$$

The probability is maximized when $x_1 = \dots = x_n = 1$, and it is minimized when one $x_i = n$ and all the other components are zero. There is one unordered sample with probability p_1 , but there are n unordered samples with probability p_2 . So:

$$ratio = \frac{p_1}{np_2} = (n-1)!$$

Problem 4

In the birthday problem, we assumed that all 365 days of the year are equally likely (and excluded February 29). In reality, some days are slightly more likely as birthdays than others. For example, scientists have long struggled to understand why more babies are born 9 months after a holiday. Let $\mathbf{p} = (p_1, p_2, \dots, p_{365})$ be the vector of birthday probabilities, with p_j the probability of being born on the j th day of the year (February 29 is still excluded, with no offense intended to Leap Dayers). The k th elementary symmetric polynomial in the variables $x_1, \dots, x_n$ is defined by

$$e_k(x_1, \dots, x_n) = \sum_{1 \leq j_1 < j_2 < \dots < j_k \leq n} x_{j_1} \dots x_{j_k}.$$

This just says to add up all of the $\binom{n}{k}$ terms we can get by choosing and multiplying k of the variables.

For example, $e_1(x_1, x_2, x_3) = x_1 + x_2 + x_3$, $e_2(x_1, x_2, x_3) = x_1x_2 + x_1x_3 + x_2x_3$, and $e_3(x_1, x_2, x_3) = x_1x_2x_3$. Now let $k \geq 2$ be the number of people.

(a) Find a simple expression for the probability that there is at least one birthday match, in terms of $\mathbf{p}$ and an elementary symmetric polynomial.

(b) Explain intuitively why it makes sense that P (at least one birthday match) is minimized when $p_j = \frac{1}{365}$ for all j , by considering simple and extreme cases.

(c) The famous arithmetic mean-geometric mean inequality says that for $x, y \geq 0$

$$\frac{x+y}{2} \geq \sqrt{xy}.$$

This inequality follows from adding $4xy$ to both sides of $x^2 - 2xy + y^2 = (x-y)^2 \geq 0$. Define $\mathbf{r} = (r_1, \dots, r_{365})$ by $r_1 = r_2 = (p_1 + p_2)/2$, $r_j = p_j$ for $3 \leq j \leq 365$. Using the arithmetic mean-geometric mean bound and the fact, which you should verify, that

$$e_k(x_1, \dots, x_n) = x_1x_2e_{k-2}(x_3, \dots, x_n) + (x_1 + x_2)e_{k-1}(x_3, \dots, x_n) + e_k(x_3, \dots, x_n)$$

show that

$$P(\text{at least one birthday match} \mid \mathbf{p}) \geq P(\text{at least one birthday match} \mid \mathbf{r})$$

with strict inequality if $\mathbf{p} \neq \mathbf{r}$, where the given $\mathbf{r}$ notation means that the birthday probabilities are given by $\mathbf{r}$. Using this, show that the value of $\mathbf{p}$ that minimizes the probability of at least one birthday match is given by $p_j = \frac{1}{365}$ for all j .

(a) answer: Let P represent the probability that there is at least one birthday match. The polynomial $e_k(p_1, \dots, p_{365})$ sums over all unordered sets of k distinct birthdays. Each such set can be assigned to the k people in $k!$ ways. Therefore,

$$P(\text{no birthday match}) = k!e_k(p_1, \dots, p_{365}),$$

and hence

$$P(\text{at least one birthday match}) = 1 - k!e_k(p_1, \dots, p_{365}).$$

$$P = 1 - k!e_k(p_1, \dots, p_{365})$$

(b) answer: If one day had birthday probability 1, then everyone would be born on that day, so

$$P(\text{at least one birthday match}) = 1.$$

More generally, if the birthday probabilities are concentrated on only a few days, then different people are more likely to be born on the same day.

For a simple example, suppose there are only two possible days, with probabilities p and $1-p$. The probability that two people have the same birthday is

$$p^2 + (1-p)^2 = 2 \left(p - \frac{1}{2} \right)^2 + \frac{1}{2},$$

which is minimized when $p = 1/2$.

Thus, making the birthday probabilities more equal reduces the chance of a birthday match. Therefore, it is intuitive that the probability of at least one birthday match is minimized when

$$p_j = \frac{1}{365}$$

for all j .

(c) answer:

(1).Proof of the equation:

$x_1 x_2 e_{k-2}(x_3, x_4, \dots, x_n)$ represents all the cases that x_1, x_2 are chosen together with the rest $k-2$ elements.

$(x_1 + x_2) e_{k-1}(x_3, \dots, x_n)$ represents all the cases that one of x_1 and x_2 is chosen together with rest $k-1$ elements.

$e_k(x_3, \dots, x_n)$ represents all the cases that none of x_1 or x_2 is chosen.

(2).Proof of the question:

$$P(\text{at least one birthday match} \mid \mathbf{p}) = 1 - k! [p_1 p_2 e_{k-2}(p_3, \dots, p_n) + (p_1 + p_2) e_{k-1}(p_3, \dots, p_n) + e_k(p_3, \dots, p_n)]$$

Since

$$r_1 = r_2 = \frac{p_1 + p_2}{2},$$

we have

$$r_1 + r_2 = p_1 + p_2$$

and

$$r_j = p_j, \quad j = 3, 4, \dots, 365.$$

Using the identity given in the problem, we obtain

$$\begin{aligned} e_k(\mathbf{r}) - e_k(\mathbf{p}) &= (r_1 r_2 - p_1 p_2) e_{k-2}(p_3, \dots, p_{365}) \\ &\quad + [(r_1 + r_2) - (p_1 + p_2)] e_{k-1}(p_3, \dots, p_{365}). \end{aligned}$$

Since $r_1 + r_2 = p_1 + p_2$, the second term is zero. Therefore,

$$\begin{aligned} e_k(\mathbf{r}) - e_k(\mathbf{p}) &= (r_1 r_2 - p_1 p_2) e_{k-2}(p_3, \dots, p_{365}) \\ &= \left(\frac{(p_1 + p_2)^2}{4} - p_1 p_2 \right) e_{k-2}(p_3, \dots, p_{365}) \\ &= \frac{(p_1 - p_2)^2}{4} e_{k-2}(p_3, \dots, p_{365}) \\ &\geq 0. \end{aligned}$$

Since all birthday probabilities are positive,

$$e_{k-2}(p_3, \dots, p_{365}) > 0.$$

Therefore, if $\mathbf{p} \neq \mathbf{r}$, then $p_1 \neq p_2$, and

$$e_k(\mathbf{r}) - e_k(\mathbf{p}) > 0.$$

Hence,

$$P(\text{at least one birthday match} \mid \mathbf{r}) < P(\text{at least one birthday match} \mid \mathbf{p}).$$

Let $(q_1, q_2, \dots, q_{365})$ be the vector that makes $P(\text{at least one birthday match} \mid \mathbf{q})$ the minimum.

If there are two probabilities q_i and q_j that are not equal, let

$$q_i^* = q_j^* = \frac{q_i + q_j}{2},$$

and replace q_i and q_j with q_i^* and q_j^* , while keeping all other components unchanged. Then $\mathbf{q}$ becomes $\mathbf{q}^*$.

According to the inequality proved above,

$$P(\text{at least one birthday match} \mid \mathbf{q}^*) < P(\text{at least one birthday match} \mid \mathbf{q}).$$

Therefore, $\mathbf{q}$ is not the vector that minimizes the probability, which is a contradiction.

Hence, $P(\text{at least one birthday match})$ has its minimum value only when all components of the vector are equal. Since their sum is 1, the value of $\mathbf{p}$ that minimizes the probability is given by

$$p_j = \frac{1}{365}$$

for all j .

Based on the above content, the statement has been proved.

